COMPANY CAPABILITY BRIEF

EchoMind

A private, on-premises AI platform for enterprise knowledge work — document intelligence, real-time speech, and conversational AI inside your own security boundary.

EchoMind is Ajace AI's enterprise AI platform. It lets an organisation ask questions of its own documents, transcribe and understand its meetings, hold a natural spoken conversation with its knowledge, and generate finished business documents — all running on a single appliance the organisation owns, with no cloud dependency and no data leaving the building.

What makes it different

Private by design

Runs fully on-premises, at the edge, or air-gapped. Model weights are on the appliance; nothing is sent to a third-party service.

Grounded and cited

Every answer is built only from your own content and carries a citation. When the answer is not in the sources, it says so rather than inventing one.

Multimodal

One platform spanning typed chat, live meeting transcription, and full-duplex voice — over a single shared knowledge base.

Multi-tenant

A single deployment can present as several domain products, each with its own isolated data, persona, and branding.

The capability portfolio

1. **Knowledge Chat** — Private retrieval-augmented Q&A with citations
2. **Live Transcript & Silent Assistant** — Real-time transcription with live fact-checking
3. **Voice Conversation** — Full-duplex spoken assistant over your knowledge
4. **Document Studio** — AI generation of reports, documents and decks
5. **Boardroom** — Multi-speaker meeting capture and analysis
6. **Vertical Solutions** — Domain products for health, law, banking, retail, meetings
7. **Secure Export Gateway** — Sensitive-data detection and redaction on the way out

Each capability is described on the pages that follow. Page 9 covers the shared technology and security foundation.

Knowledge Chat

Ask questions in plain language and get answers built only from your own documents and transcripts — with a citation on every claim.

Overview

Knowledge Chat is a private question-answering assistant over an organisation's own content. Users type a question and receive a direct, written answer assembled strictly from uploaded documents and stored meeting transcripts. Every statement carries a citation to its source document, section and page, and when the answer is not present in the sources the assistant says so rather than guessing.

How it works

The question is answered by a governed retrieval pipeline. Two searches run together — a semantic search over meaning (vector embeddings) and an exact keyword search — and their results are fused and ranked. A cross-encoder then re-reads the strongest passages against the question and keeps the best. A relevance gate drops weak context rather than forcing it in. Finally the language model writes the answer from that evidence alone, attaching citations.

Key technology

Retrieval	Hybrid FAISS vector + BM25 keyword search, weighted rank fusion
Re-ranking	Cross-encoder relevance re-scoring of top candidates
Generation	On-device open-weight LLM, answer grounded in retrieved passages
Guardrail	Abstains when no relevant source is found

Business value

- Verifiable answers — every claim is traceable to a source
- No hallucinated citations; abstains rather than inventing
- Turns scattered documents into an answerable knowledge base
- Works fully offline on private and regulated content

Representative use cases

Employee policy and HR self-service, contract and regulation look-up, technical support, onboarding, research and analysis across large document sets.

PROOF POINT In a measured evaluation, Knowledge Chat achieved 0.98 citation precision and abstained on 78% of genuinely unanswerable questions, with zero fabricated citations.

CAPABILITY 2 · SPEECH & COMPLIANCE

Live Transcript & Silent Assistant

Transcribe meetings and calls in real time — while a background AI silently checks each statement against your knowledge base.

Overview

This capability turns live speech into an accurate, searchable record and adds a layer of real-time assurance on top of it. As people speak, their words appear on screen; when the meeting ends the transcript is stored and becomes answerable alongside documents. Running quietly alongside, the Silent Assistant evaluates each completed statement against reference material and labels it — supported, contradicted, unverified, or a compliance risk.

How it works

Audio streams to the appliance in small frames over a live connection. A streaming speech model produces captions continuously, and natural pauses close paragraphs. Each finished paragraph is passed through the same retrieval stack used by Knowledge Chat and judged by the language model; domain rule packs sharpen the check per industry. Only findings above a confidence threshold are surfaced, so the room is not flooded with noise.

Key technology

Live ASR	NVIDIA NeMo streaming speech-to-text over a real-time link
Checking	Each paragraph retrieved and judged against the knowledge base
Domains	Per-industry rule packs (clinical, legal, banking, retail)
Capture	Transcripts auto-saved and made searchable

Business value

- Catch an incorrect or non-compliant statement while it can still be corrected
- Automatic, searchable record of every meeting
- Real-time compliance and quality assurance
- Yesterday's discussion answers today's question

Representative use cases

Clinical consultations, financial advice and KYC calls, legal discussions, sales conversations, board and project meetings, and any setting where what is said must align with policy.

Voice Conversation

A natural spoken conversation with your knowledge base — talk, get answers out loud, and interrupt mid-sentence.

Overview

Voice Conversation is a hands-free, full-duplex spoken assistant. The user simply talks; the assistant answers aloud from the same documents and transcripts that power Knowledge Chat, and can be interrupted at any moment, just like a person. It is designed to feel responsive and natural rather than the stilted, wait-for-the-beep experience of typical voice bots.

How it works

The system continuously listens and decides when a turn has ended using voice-activity detection and semantic end-pointing — a finished sentence gets a fast reply, a trailing one is given more time. A short spoken lead phrase begins within a fraction of a second to cover the thinking time, while the fully grounded answer is retrieved and streamed into speech behind it. If the user starts talking, playback stops immediately and the system listens again.

Key technology

Turn-taking	Voice-activity detection + semantic end-pointing
Responsiveness	Immediate spoken lead phrase; grounded answer streams behind
Barge-in	Playback halts the instant the user speaks
Accuracy	Second-pass ASR feeds the exact transcript to the model

Business value

- Natural, low-latency spoken interaction
- Hands-free operation for field, clinical and mobile use
- Accessibility for users who cannot type
- Same grounded, cited knowledge as the typed assistant

Representative use cases

Field maintenance and inspections, clinical dictation and bedside queries, driving or warehouse use, and accessible access to enterprise knowledge.

PROOF POINT In testing, the assistant began speaking in about 0.6 seconds and reliably stopped on interruption.

CAPABILITY 4 · CONTENT GENERATION

Document Studio

Turn a short brief into a finished, formatted business document or presentation — grounded in your own content.

Overview

Document Studio generates complete, ready-to-send deliverables rather than chat text to be copied and reformatted. From a short brief it produces polished PDF and PowerPoint files across a library of business templates — clinical notes, contract-review memos, meeting minutes, whitepapers, proposals, pitch decks, case studies and more — each in a consistent, branded visual theme.

How it works

Each template provides a section blueprint and writing guidance. The language model drafts each section, grounded in the supplied brief or the organisation's knowledge base, and a rendering engine produces a genuine PDF or PowerPoint file — not a screenshot or a text dump. Supporting illustrations are generated on-device. Organisations can add their own custom templates.

Key technology

Templates	23 business document and presentation templates, 5 visual themes
Drafting	LLM section-by-section, grounded in brief or knowledge base
Output	Real PDF and PowerPoint files
Imagery	On-device image generation (SDXL-Turbo)

Business value

- Hours of drafting and formatting reduced to minutes
- Consistent, on-brand output every time
- Deliverables, not transcripts — ready to send
- Everything generated privately on the appliance

Representative use cases

Clinical documentation, contract review memos, meeting minutes, sales proposals and quotes, internal reports, marketing and training material, and executive presentations.

Boardroom

Record a multi-speaker session, separate who said what, and produce an analysed, exportable report.

Overview

Boardroom captures multi-person meetings end to end. It records the session, distinguishes between speakers, links the audio to a written transcript, and produces an analysed report of what was decided and what needs to happen next — replacing the manual work of writing minutes.

How it works

Audio is captured and uploaded in segments, then assembled and normalised into a standard format. The system separates the recording into speaker turns, links it to the live transcript, and analyses the content into decisions, commitments and action items, which can be exported for distribution.

Key technology

Capture	Segmented recording, assembled and normalised on-device
Diarisation	Separation of the conversation into speaker turns
Analysis	Decisions, commitments and action items extracted
Output	Structured, exportable meeting report

Business value

- Minutes and action items produced automatically
- A reliable record of who committed to what
- Decisions captured instead of lost
- Private capture of sensitive discussions

Representative use cases

Board and committee meetings, project stand-ups, client and stakeholder calls, interviews, and any multi-party discussion that needs a formal record.

Vertical Solutions

One platform presenting as several domain products — each with its own isolated knowledge, persona and branding.

Overview

The same EchoMind deployment can operate as several distinct, industry-specific products at once — for health, law, banking, retail and meetings. Each presents its own knowledge base, subject-matter persona, terminology and visual identity, while sharing the underlying engine and security controls. This lets a single appliance serve different departments, or a single product be sold into different regulated industries, without their data ever mixing.

How it works

Each vertical maps to an isolated namespace. Crucially, that isolation is applied inside the search itself rather than as a filter after the fact — a query in one vertical cannot retrieve another vertical's content even during ranking. Each vertical also carries its own persona, domain rule packs, and theme.

Key technology

Isolation	Namespace enforced inside the retrieval query, not post-filtered
Personas	Subject-matter persona and tone per vertical
Compliance	Domain-specific rule packs (e.g. clinical, KYC/AML)
Branding	Independent theme and terminology per product

Business value

- One appliance serves many departments or customers
- Guaranteed data separation between tenants
- Faster go-to-market across regulated industries
- Domain-tuned behaviour without separate builds

Representative use cases

Hospitals and clinics, law firms, banks and lenders, retail and sales teams, and shared corporate deployments serving multiple business units.

PROOF POINT In isolation testing, cross-tenant queries produced zero leaks across every retrieval path checked.

Secure Export Gateway

Before any content leaves the appliance, it is scanned for sensitive data and a redacted version is produced.

Overview

The Secure Export Gateway is a data-loss-prevention layer for AI output. Whenever content is exported from the private boundary, it is first inspected for sensitive information, which is ranked by severity and masked. It is fully deterministic and runs offline, so it can be applied on every export — including in air-gapped deployments — without a GPU or network connection.

How it works

The gateway applies a set of precise detectors for sensitive data types — national identifiers, payment card numbers (with a mathematical validity check), API keys, private keys, security tokens, email addresses, phone numbers and network addresses. Each match is classified by severity and masked, and the gateway returns both the risk summary and a safe, redacted copy of the content.

Key technology

Detection	National IDs, card numbers, keys, tokens, emails, phones, IPs
Validation	Luhn check to confirm genuine card numbers
Operation	Deterministic, offline — no GPU or network needed
Output	Severity ranking plus a redacted copy

Business value

- “Nothing leaves unchecked” as an enforced mechanism, not a policy
- Safe bridge from an offline system to the outside world
- Runs on every export, including air-gapped
- Supports data-protection and privacy obligations

Representative use cases

Releasing summaries or reports outside a secure boundary, sharing AI output with external parties, and any offline-to-online hand-off of regulated or confidential material.

Technology & Security Foundation

The common engine and controls beneath every EchoMind capability.

On-device AI stack

Every model runs locally on a single NVIDIA DGX Spark appliance (GB10 Grace-Blackwell, 128 GB unified CPU-GPU memory). No inference leaves the machine.

Model and retrieval components

Language model	On-device open-weight large language model (30B-class, mixture-of-experts, 4-bit NVFP4 quantization) served by NVIDIA TensorRT-LLM behind an OpenAI-compatible endpoint. The platform is model-agnostic — the model can be changed with a single setting.
Speech-to-text	NVIDIA NeMo — a streaming model for live captions and Parakeet-TDT for the accurate final transcript.
Text-to-speech	Piper neural voices for spoken output.
Embeddings	nomic-embed-text (768-dimension) for semantic search.
Image generation	SDXL-Turbo for on-device document artwork.
Retrieval	FAISS dense vectors + BM25 keyword index, fused and re-ranked by a cross-encoder; SQLite for metadata and audit.

Security and governance

- No runtime egress — the system operates with the network cable unplugged; model weights are baked into the appliance.
- Tenant isolation is enforced inside the retrieval query itself, not by filtering results afterwards, so one tenant's data cannot surface in another's ranking.
- Retrieved text is treated as evidence, never as instructions — a document cannot hijack the assistant (prompt-injection defense).
- Role-based access control, per-user tenant scoping, and an audit log of uploads, queries and exports.
- The Secure Export Gateway scans any outbound content for sensitive data and returns a redacted version.
- Optional public access is an outbound-only, identity-gated tunnel — no inbound ports are opened.

Engineering rigour

The platform ships as containerised services started with a single command. A 52-question automated quality suite gates every release. The architecture and a measured evaluation of the system have been accepted for publication at an international peer-reviewed conference (QASC 2026), covering latency, grounding fidelity, tenant isolation, and adversarial robustness.